FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2019

PAPER – 1

TEST DATE: 13-05-2019

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

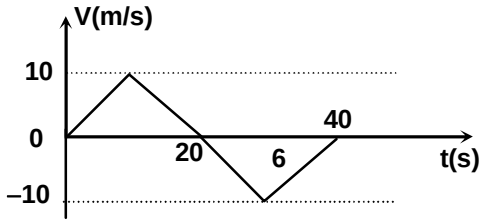
Physics

PART – I

SECTION – A
(One or More than one correct type)

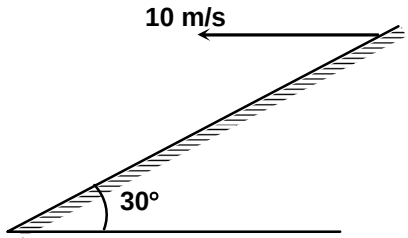
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Velocity time graph of a particle moving along straight line is shown in figure. Choose correct option(s).



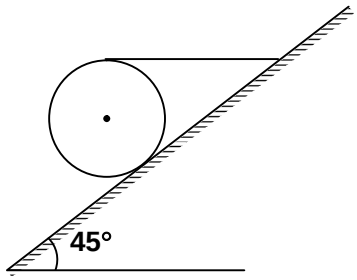
(A) Average velocity of the particle through out motion is zero.
(B) Average speed of the particle is 5 m/s.
(C) Distance travelled by the particle in 40 sec is 200 m.
(D) Displacement of the particle during first 20 sec is 100 m.

2. A particle is projected horizontally from an inclined plane with speed 10 m/s as shown in figure. ($g = 10 ms^{-2}$) choose correct option(s)



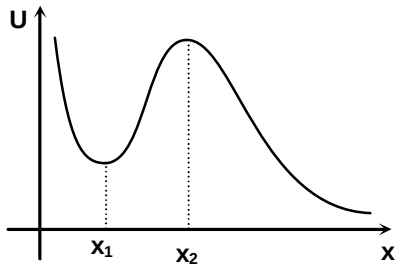
(A) Range along inclination is 10 m
(B) Time of flight is $\frac{2}{\sqrt{3}}$ sec
(C) Speed of particle when it strikes inclined plane is 15 m/s
(D) Time of flight is $\sqrt{3}$ sec.

3. A sphere of mass 10 kg is kept stationary on a rough inclined plane as shown in figure. String is massless and inextensible. At equilibrium string remains horizontal. Choose correct statement(s) : ($g = 10\ ms^{-2}$)



(A) Tension in string is $\frac{100}{\left(\sqrt{2}+1\right)}$N
(B) Friction force acting on sphere is $\left(\frac{100}{\sqrt{2}+1}\right)$N
(C) Minimum value of coefficient of friction for equilibrium is 1
(D) Normal reaction between sphere and inclination is $50\sqrt{2}$ N.

4. The given plot shows the variation of potential energy of interaction between two particles with distance between them. (Particles are placed in isolated space).



(A) Mutual force acting on the particles is zero when distance between then is x_1
(B) Mutual force acting on the particles is zero when distance between then is x_2
(C) Mutual force between the particles is repulsive when distance between them is more than x_1 but less than x_2.
(D) Equilibrium is stable at x_1.

5. Electrons having de-Broglie wave length λ are incident on target in X-ray tube. Minimum wavelength of X-rays generated is λ_1. Then choose INCORRECT statement(s).
(A) If λ is increased the value of λ_1 decreases.
(B) If λ is increased the value of λ_1 increases.
(C) The value of λ_1 will not change if target metal is changed.
(D) The value of λ_1 does not depend on λ.

6. Two waves having same wavelength λ in vacuum are in phase initially. First wave travels a path length L_1 in medium of refractive index n_1 and second wave travels path length L_2 in medium of refractive index n_2. They are now combined in vacuum. Based on above information choose correct option(s).
(A) Phase difference between waves after combination is $\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$.
(B) Frequency of both waves remains unchanged when they travel through different medium.
(C) Phase difference between waves after combination is $\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$
(D) Frequency of wave in medium with more refractive index will be more.

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8

A circular loop of radius R is kept in x–y plane with centre at origin. Magnetic field in region is given by $\vec{B} = B_0\hat{i} + 2B_0\hat{j} + 3B_0\hat{k}$. Based on above information, answer the following questions.

7. If loop starts rotating with angular velocity ω about x–axis. Then value of average induced emf during the time loop changes its plane from x-y plane to x–z plane for the first time is
(A) $2B_0R^2\omega$
(B) $\frac{2B_0\omega R^2}{\pi}$
(C) $B_0\omega R^2$
(D) $\frac{B_0\omega R^2}{\pi}$

8. If a current i is flowing in loop when it was in x–y plane and at rest. The direction of current is such that when an observer situated at positive z-axis looks at loop he finds direction of current clockwise. The value of net torque about centre of loop is

(A) $2i\pi R^2 B_0\left(\hat{i}-\frac{\hat{j}}{2}\right)$

(B) $i\pi R^2 B_0\left(\hat{i}-2\hat{j}\right)$

(C) $2i\pi R^2 B_0\left(\hat{i}+\frac{\hat{j}}{2}+\hat{k}\right)$

(D) $i\pi R^2 B_0\left(\hat{i}+2\hat{j}+\hat{k}\right)$

Paragraph for Question Nos. 9 and 10

An experiment is performed to calculate the focal length of a convex mirror using convex lens. The setup consists of two pins, optical bench, one convex lens and one convex mirror.
When only convex lens is placed at a distance 60 cm from pin (object) its image is formed at a distance 30 cm from the lens as shown in figure (i) Now. Convex mirror is also kept right to the lens without changing distance between pin (object) and lens. The mirror is kept at a distance 10 cm from lens such that position of image of object pin and position of object pin coincides as shown in figure (ii)

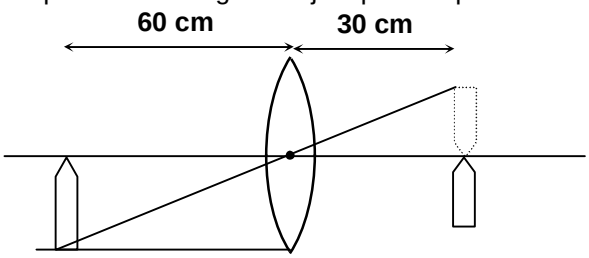


Figure (i)

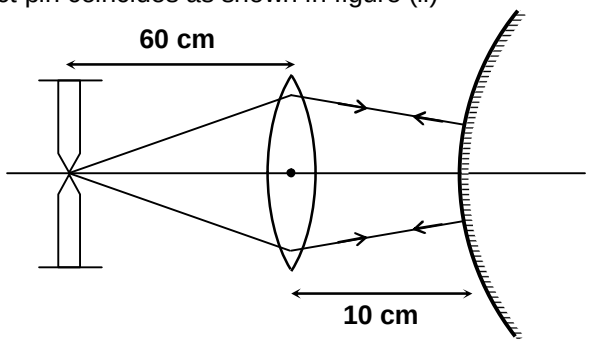


Figure (ii)

Now answer the following questions.

9. Focal length of convex lens used is
(A) 20 cm
(B) 40 cm
(C) 60 cm
(D) 30 cm

10. Focal length of convex mirror is
(A) 10 cm
(B) 30 cm
(C) 20 cm
(D) 60 cm

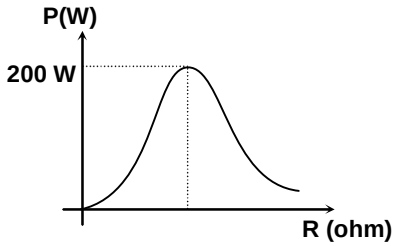
SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A uniformly charged ring is placed in x – y plane with its centre at origin. Total charge on ring is (10 nC) and radius of ring is 1 cm. The magnitude of potential difference between points having co–ordinate (0, 0, $2\sqrt{2}$ cm) and (0, 0 $-2\sqrt{6}$ cm) is (in volts)

12. The magnetic field in a certain cylindrical region of radius 5 cm is varying with time according to law given by B = ($4t^2 - 3t$) where B is in Tesla and t is in second. The value of induced electric field at 3 cm from centre is E_1 and at 6 cm from centre is E_2. The value of product of magnitude of E_1 and E_2 (at t = 1 sec) is $k \times 10^{-5}$ V^2/m^2. The value of k is

13. A battery of emf 48 V unknown internal resistance is connected to a rheostat whose resistance can be varied in a long range. A graph is plotted between power developed across rheostat and resistance of rheostat which is shown in figure. The internal resistance of the battery in Ω is



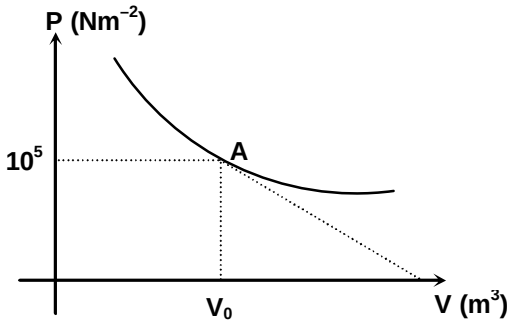
14. Moment of inertia of a uniform equilateral triangular lamina of mass 20 kg and side length 75 cm about an axis perpendicular to plane and passing through centroid (in kg–m^2) is I. The value of 100 I is

15. If pressure inside a spherical bubble of radius 0.2 mm just below the surface of water is $k \times 10^4$ N/m^2 then the value of k is …….. (Given surface tension of water = 7×10^{-2} N/m and atmospheric pressure 1.013×10^5 N/m^2)

16. Gravitational field intensity at the centre of the base of a uniform hollow hemisphere of mass 2 kg and radius 50 cm is $E \times 10^{-11}$ N/kg. The value of E is

17. A harmonic pulse given by $y(\text{in cm}) = (0.9 \text{ cm})\sin\left(t - \frac{x}{4}\right)$. x in meter and t in second is travelling in a uniform tight string. Same pulse is transmitted onto another string where speed of wave becomes 2 m/s. The amplitude of transmitted wave in (cm) is

18. P-V graph of an ideal di-atomic gas undergoing adiabatic process is shown in figure. At certain point A, tangent is drawn slope of tangent is -1 (N/m^5). The volume of the gas at point A is $V \times 10^4$ m^3. The value of V is



Chemistry PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

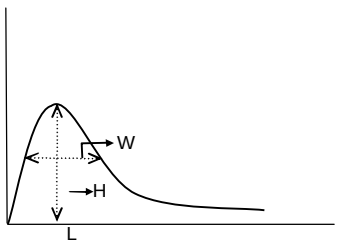
19. Which of the following statement(s) regarding alkaline earth metals is/are correct?
(A) Only BeF_2 behaves as a Lewis acid while other fluorides of alkaline earth metals don't act as Lewis acid.
(B) Be doesn't react directly with fluorine hence BeF_2 is prepared by heating $(NH_4)_2BeF_4$.
(C) In solid state the geometry of $BeCl_2$ is tetrahedral.
(D) All metals of alkaline earth metals show characteristic colour in flame test.

20. Which of the following complex(es) is/are diamagnetic as well as show(s) optical isomerism
(A) $[PtCl_2 (MeNH –CH_2 – CH_2 – NHMe)]$
(B) $\left[Co\left(C_2O_4\right)_3\right]^{3-}$
(C) $K\left[Cr\left(H_2O\right)_2\left(C_2O_4\right)_2\right]$
(D) $\left[Ni\left(en\right)_2 Cl_2\right]$

21. Which of the following halogen react(s) with water at room temperature to give hypohalous acid (HOX)
(A) F_2
(B) Cl_2
(C) Br_2
(D) I_2

22. The correct statement(s) pertaining to colloids is/are
(A) The process of converting freshly prepared precipitate into colloidal sol by shaking it with dispersion medium in the presence of excess amount of electrolyte is called peptization.
(B) Lyophilic colloids are also called as reversible sols.
(C) Sulphur sol is an example of multimolecular colloid.
(D) Sols are those colloids in which dispersion medium is liquid phase while dispersed phase is solid phase.

23. The Maxwell-Boltzmann distribution for molecular speeds of a gas is shown in the figure at given temperature T



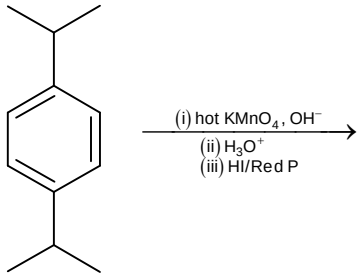
In the figure H is the height of the peak, L is the location of the maximum and W is the width at half height. Which of the following statement(s) is/are correct regarding this graph:
(A) For a given gas, as the temperature decreases, H decreases, L increases and W increases.
(B) At a given temperature as the molar mass decreases H increases, L decreases and W decreases.

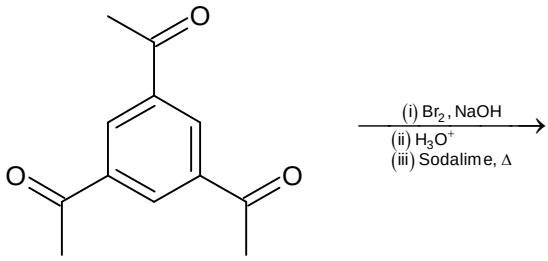
(C) For a given gas, as the temperature decreases, H increases, L decreases and W decreases.

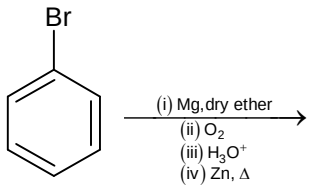
(D) At a given temperature as the molar mass decreases H decreases, L increases and W increases.

24. The reaction(s) leading to the formation of Benzene is/are

(A) $n-hexane \xrightarrow[773\ K,\ 10-20\ atm]{Cr_2O_3 \text{ or } V_2O_5}$

(B) 

(C) 

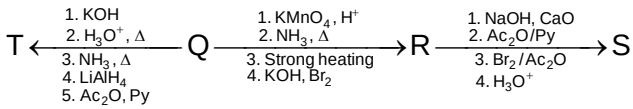
(D) 

(Paragraph Type)

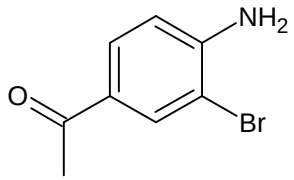
This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

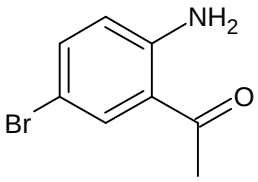
Paragraph for Question Nos. 25 and 26

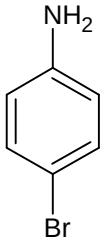
A hydrocarbon 'P' (C_8H_{10}) on oxidation gives a dicarboxylic acid which reacts with ethylene glycol produce a polymer Glyptal. The compound 'P' on reaction with CrO_3/Ac_2O followed by acidic hydrolysis give a compound Q. Q undergoes the following sequences to furnish S via R. The compound Q also undergoes another set of reaction to produce T.

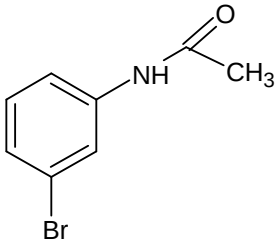


25. The compound S is

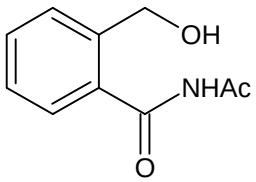
(A) 

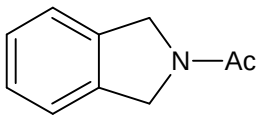
(B) 

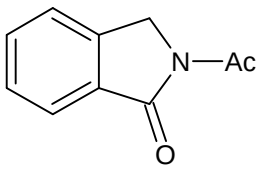
(C) 

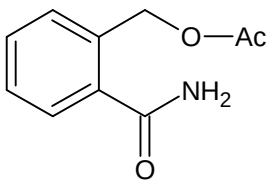
(D) 

26. The compound T is:

(A) 

(B) 

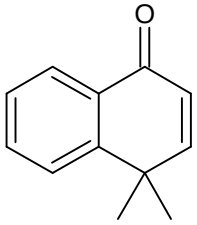
(C) 

(D) 

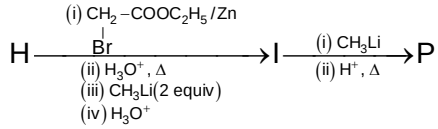
Paragraph for Question Nos. 27 and 28

A 3° alcohol P upon acid catalysed dehydration gives a product Q. Ozonolysis of Q leads to only compound R which upon heating with KOH gives a compound S.

S =



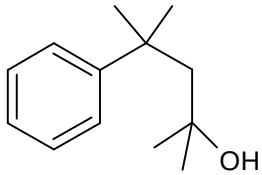
3° alcohol 'P' is synthesized by the following route:



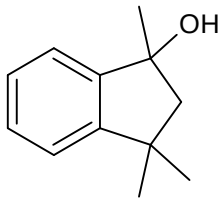
Molecular formula of H is C_8H_8O and it gives positive iodoform test.

27. The compound 'P' is:

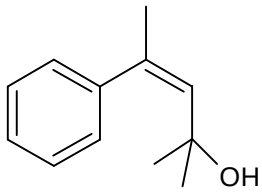
(A)



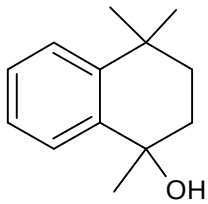
(B)



(C)

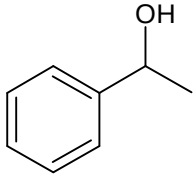


(D)

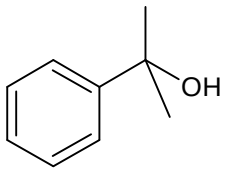


28. The compound 'H' on reaction with CH_3MgBr followed by hydrolysis gives:

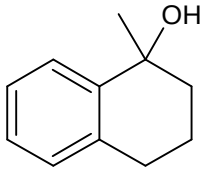
(A)



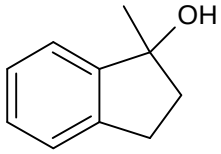
(B)



(C)



(D)



SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

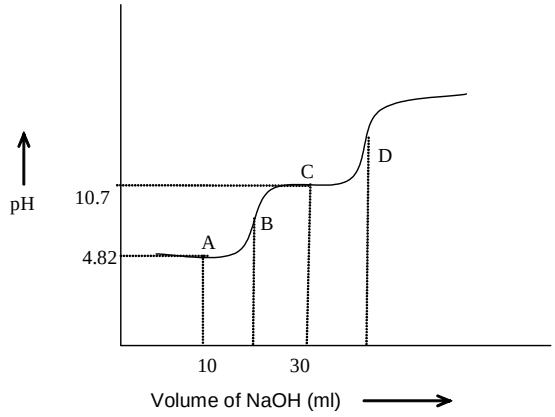
29. The density of solid XY having Rock salt type structure is 90 mg/cm^3. If the distance between X and Y is $n \times 10^{-10}$ meter. The n is (given atomic mass of X and Y are 42 and 66 gm mol^{-1} respectively and $N_A = 6 \times 10^{23}$)

30. The equilibrium of the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

is set up at 300 K by mixing N_2(g) and H_2(g). If the standard free energy of formation of NH_3(g) is – 15.53 kJ/mol. If after one hour of mixing the partial pressure of N_2, H_2 and NH_3 are 50, 2 and 200 bar respectively and the enthalpy change of reaction is – 4.2 kJ, what is the entropy change of the reaction. (in J K^{-1}) [Given: $\ell n 100 = 4.6, RT = 2500 \text{ JK}^{-1}$]

31. 4-amino butanoic acid is an important neurotransmitter. The plot given below shows the titration curve for 20 ml of 0.1 M of 4-aminobutanoic acid hydrochloride against 0.1 M of NaOH



The pH of the solution when 20 ml of 0.1 M of 4-aminobutonoic acid hydrochloride is titrated with 25 ml of 0.1 M NaOH solution is [Given: $\log_{10} 3 = 0.48$]

32. The resistance of $Ba(OH)_2$ is 4000 times greater than KCl if measured in the same conductivity cell. The molar conductivity of $BaCl_2$, NaOH and NaCl at infinite dilution are 2x, 4x and 3x $Scm^2 mol^{-1}$ respectively. If the degree of dissociation of 10^{-2} M $Ba(OH)_2$ is a × 10^{-2}. The value of 'a' is: [Given: specific conductivity of KCl is 0.002x Scm^{-1}]

33. The rate expression for the following reaction

$A(g) \longrightarrow 2B(g) + C(s) + D(g)$

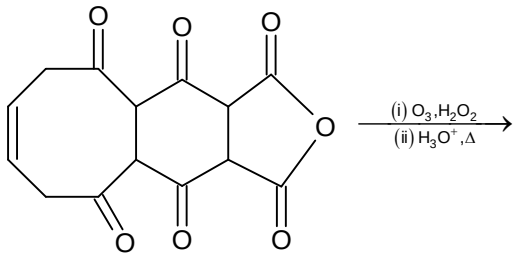
is $r = k[A]$ at 500 K and at constant volume.

The pressure of system after 10 second and after a very long time was found to be 21 and 27 bar respectively. If the conc. of A after 20 second was 2 mol/litre, then rate of reaction (in mol. $Litre^{-1}$ $minute^{-1}$) after 20 minute is [Given $\ell n3 = 1.1$]

34. The solubility product of AgBr is 1×10^{-13} at 298 K. If standard reduction potential of half cell, $E^{o}_{Ag/AgBr(s)/Br^-}$ is 0.0725 V, the standard reduction potential, $E^{o}_{Ag^+/Ag}$ (in volt) is:

$\left[\text{Given}: \frac{RT}{F} = 0.025,\ \ell n10 = 2.3\right]$

35. Number of carbonyl groups present in the final product of the following reaction sequence is P then $\frac{P}{5}$ is:



36. Total number of compounds in which central atom have only two lone pair is x then $\frac{x}{2}$ is:

$ClF_4^+, SeF_2, XeOF_2, XeO_3, ClF_2^+, SNF_3, I_3^-, SOCl_2, SO_3^{2-}$

Mathematics PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. In a ΔABC, if all angles are less than $\frac{2\pi}{3}$ and $S = \frac{\cos A + \cos B - \cos C}{\sin A + \sin B - \sin C}$, then which of the following values of S is/are possible

(A) $-\frac{1}{\sqrt{3}}$

(B) $-\frac{1}{2}$

(C) $\frac{1}{\sqrt{3}}$

(D) $\frac{1}{2}$

38. Consider a pyramid built on a square base of side 2 such that the lengths of the line segments joining its apex to the 4 corners of the base are equal and the height of the structure is equal to the length of a side of its base. Then which of the following statements is/are true

(A) volume of the pyramid is $\frac{8}{3}$

(B) the angle between the base and any other face is $\sin^{-1}\frac{2}{\sqrt{5}}$

(C) the angle between two faces with common edge which passes through the apex is $\cos^{-1}\frac{1}{5}$

(D) volume of the largest tetrahedron that can be inscribed in the pyramid is 1

39. Consider the function $F(x) = \int_1^n \frac{x^u}{u}du$ (n > 1), then which of the following statements is/are true

(A) F(x) has a local maxima at x = 2

(B) $\lim_{x\to 1^+} F(x) = 0$

(C) $\lim_{x\to 1} F(x) = \ln n$

(D) $\lim_{x\to 1^+} F(x) = \ln n$

40. Two functions f(x) and g(x) are defined as $f(x) = \begin{cases} [x]^{\{x\}} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ and $g(x) = \begin{cases} \{x\}^{[x]} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$, (where [.] and {.} refer to the greatest integer function and the fractional part function respectively). Then which of the following statements is/are true (where $n \in N$)

(A) $\int_0^{n+1} f(x)dx - \int_0^n f(x)dx = \frac{n-1}{\ln n}$

(B) $\int_0^{n+1} g(x)dx - \int_0^n g(x)dx = \frac{1}{n+1}$

(C) $\int_0^t f(x)dx = \int_0^t g(x)dx$ has at least one solution in $t \in \left(0, \frac{\pi}{2}\right)$

(D) $e < \lim_{n\to\infty} \int_0^n g(x)dx < e^2$

41. Consider $f(x) = \sin^{-1}(\sin x) - \cos^{-1}(\cos x)$, then which of the following statements is/are true

(A) $f(x) = 0$ for $x \in \left(0, \frac{\pi}{4}\right)$

(B) $f(x) < 0$ for $x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

(C) $f(x) > 0$ for $x \in \left(\pi, \frac{5\pi}{4}\right)$

(D) $f(x) < 0$ for $x \in \left(-\pi, \frac{-3\pi}{4}\right)$

42. Consider a mirror made out of hyperbola $x^2 - y^2 = 2$ such that its exterior surface is reflecting. A ray of light passes along $x - y = 2$ from the exterior of the hyperbola, then the reflected ray travels along which of the below lines.

(A) $3y - x + 3 = 0$
(B) $3y + x = 0$
(C) $y + 7x + 2 = 0$
(D) $7y + x + 2 = 0$

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Finding the area or the length of boundary of the shape enclosed by a set of curves requires the use of definite integration between limits defined by their points of intersection. Quite often it is easier to do the integration using polar coordinates

43. The area enclosed by the curve $x^3 + y^3 = 3xy$ is

(A) $\frac{3}{2}$

(B) $\frac{1}{2}$

(C) $\frac{\pi}{2}$

(D) $\frac{3\pi}{2}$

44. The length of the curve defined by $r = \theta$, where $0 \le \theta \le 1$, is

(A) $\frac{1}{\sqrt{2}}$

(B) $\frac{\pi}{\sqrt{2}}$

(C) $\frac{1}{\sqrt{2}}+\ln\sqrt{1+\sqrt{2}}$

(D) $\frac{\pi}{\sqrt{2}}+\ln\left(1+\sqrt{2}\right)$

Paragraph for Question Nos. 45 and 46

Let ABC be a triangle such that origin is its circumcentre & its base, BC, always passes through the point (4, 5). If the lateral sides (AB and AC) have x + y = 0 and x – 9y = 0 as their perpendicular bisectors, then the locus of A is a circle Γ.

45. Γ passes through which of the below points
(A) (0, 0)
(B) (1, 41)
(C) $\left(\frac{4}{5},\frac{3}{5}\right)$
(D) (–1, –2)

46. The centre of Γ lies on the line
(A) x + 5y = 7
(B) x – y = 0
(C) x = 0
(D) y = 0

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. If P is the probability of choosing two squares on a 8×8 standard chess-board such that two bishops on these squares cannot attack each other in one move, then $\left[\frac{1}{P}\right]$ is (where [.] denotes the greatest integer function).

48. Let x, y, z be three distinct complex numbers such that xyz = 5 and $\begin{vmatrix} x & x^2 & 1-yz \\ y & y^2 & 1-zx \\ z & z^2 & 1-xy \end{vmatrix} = 0$, then $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is equal to

49. Let A be a 3×3 orthogonal matrix and λ be a non-zero, real number such that $|\text{adj}(\text{adj}\ \lambda A)| = 16|\text{adj}\ \lambda A|$, then $[\lambda^2]$ is equal to (where [.] denotes the greatest integer function)

50. Let $L=\int_0^1 \lim_{n\to\infty}\sum_{k=0}^{n}\frac{x^{k+1}3^k}{k!}dx$, then $\left[\ln\frac{9}{2}L\right]$ is equal to (where [.] denotes the greatest integer function)

51. An unbiased dice with numbers {1, 2, 3, 4, 5, 6} on its faces is rolled once. An observer, A, sees the number shown on the dice and reports it. The probability that A speaks the truth is $\frac{1}{3}$ and the probability that he lies is $\frac{2}{3}$. Assume that the number he reports is always from the set {1, 2, 3, 4, 5, 6}. If p is the probability that the actual number on the dice is even given that A reports that the number was 2, then p is equal to

52. If $a_n = \sum_{r=0}^{n} \frac{1}{{}^{n}C_r}$ and $L = \lim_{n\to\infty} \frac{1}{a_n}$, then L is equal to

53. If $S = \sum_{k=1}^{\infty} \frac{1}{{}^{k+51}C_k}$, then S is equal to

54. A parallelogram is drawn to circumscribe the curve $x^2 + 4y^2 = 4$ such that two of its vertices are $\left(\sqrt{3}, \sqrt{2}\right)$ and $\left(-\sqrt{3}, -\sqrt{2}\right)$. If L is the sum of distances of the other two vertices from the centre of the curve then $\frac{1}{L^2}$ is equal to